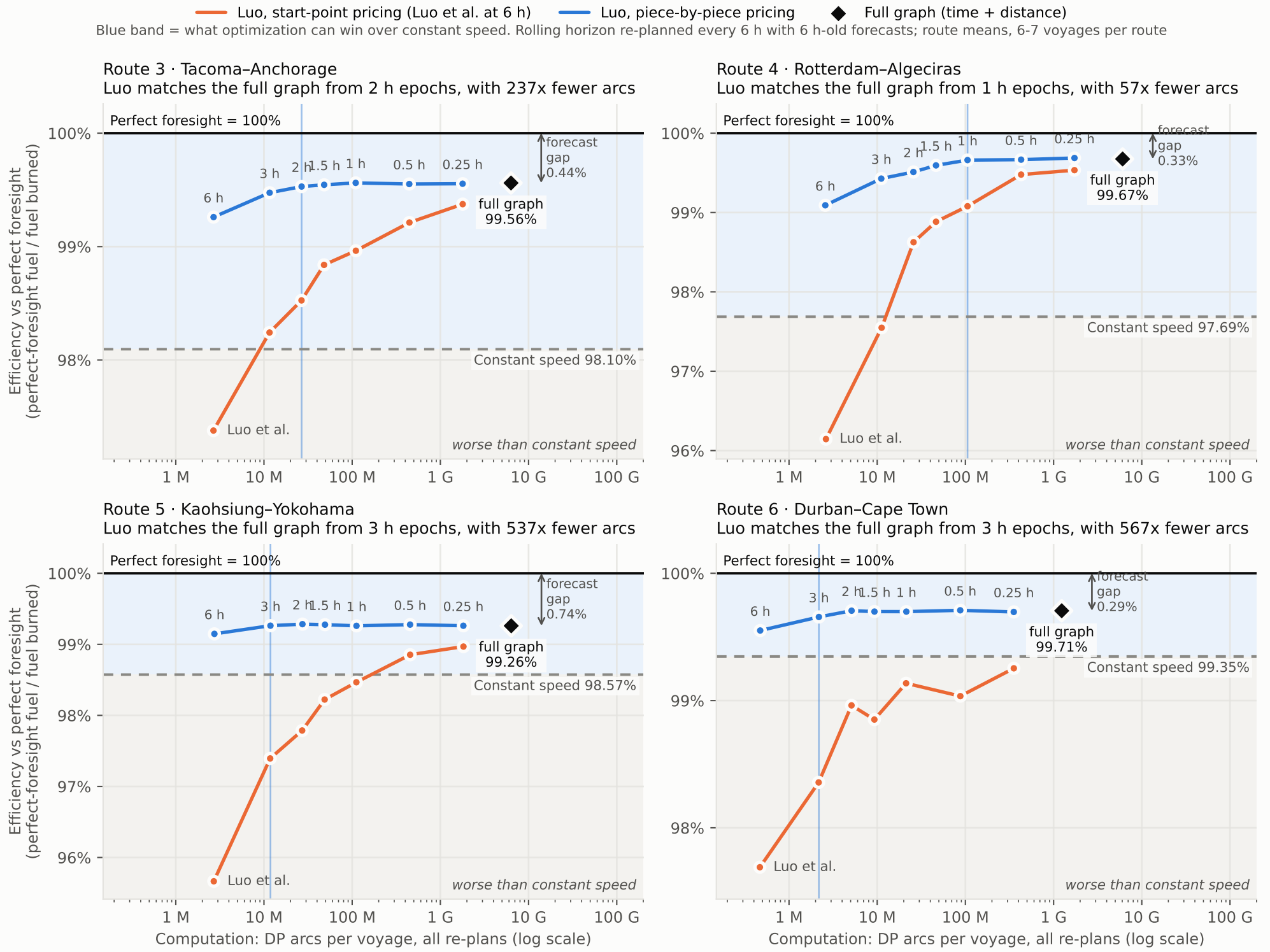
Luo, start-point pricing (Luo et al. at 6 h)
Luo, piece-by-piece pricing
Full graph (time + distance)
Blue band = what optimization can win over constant speed. Rolling horizon re-planned every 6 h with 6 h-old forecasts; route means, 6-7 voyages per route
Route 3 · Tacoma–Anchorage
Luo matches the full graph from 2 h epochs, with 237x fewer arcs
Route 4 · Rotterdam–Algeciras
Luo matches the full graph from 1 h epochs, with 57x fewer arcs
Route 5 · Kaohsiung–Yokohama
Luo matches the full graph from 3 h epochs, with 537x fewer arcs
Route 6 · Durban–Cape Town
Luo matches the full graph from 3 h epochs, with 567x fewer arcs
Perfect foresight = 100%
forecast gap 0.44%
full graph 99.56%
Constant speed 98.10%
forecast gap 0.33%
full graph 99.67%
Constant speed 97.69%
forecast gap 0.74%
full graph 99.26%
Constant speed 98.57%
forecast gap 0.29%
full graph 99.71%
Constant speed 99.35%
6 h
3 h
2 h
1.5 h
1 h
0.5 h
0.25 h
Luo et al.
worse than constant speed
Efficiency vs perfect foresight (perfect-foresight fuel / fuel burned)
Computation: DP arcs per voyage, all re-plans (log scale)
1 M
10 M
100 M
1 G
10 G
100 G